A Practitioner's Companion

for Chartered Accountants

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1 Introduction

The **GST Toolkit** is a desktop application built in Python for the day-to-day needs of Chartered Accountants, tax consultants and compliance teams. It bundles four specialised modules into a single, offline-first interface so that practitioners can validate, reconcile and export GST data without depending on external cloud services.

What's inside

GSTIN Validator

Offline Mod-36 checksum validation with automatic reverse suggestion of corrected GSTINs.

HSN / SAC Search

Fast fuzzy search by code or by description across the entire HSN and SAC master.

GSTR-1 JSON Check

Pre-upload validation of supplier GSTINs and item HSN/SAC codes, with rectification & Excel export.

GSTR-2B Reconciliation

Match purchase register against GSTR-2B B2B data, flag mismatches and export Excel/Word/PDF.

i The toolkit runs entirely on your machine. No data is transmitted anywhere — an important consideration when working with client GST data.

2 Installation & Requirements

Prerequisites

- **Python 3.9 or later** (3.10+ recommended)
- **Operating system:** Windows 10/11, macOS 12+, or modern Linux
- **RAM:** 4 GB minimum; 8 GB recommended
- Internet is required only for the first *pip install*

Install Python packages

Open a terminal / command prompt and run:

```
pip install PyQt6 openpyxl rapidfuzz pandas python-docx reportlab xlsxwriter
```

Files required

File	Purpose	Required?
gst_toolkit.py	The main application script	Yes
HSN_SAC (3).xlsx	HSN/SAC master data (loaded once, then cached)	Yes (one-time)
hsn_cache.json	Auto-generated cache — created after first load	Auto

* Place gst_toolkit.py and HSN_SAC (3).xlsx in the same folder, then run: python gst_toolkit.py

3 Getting Started

When you launch the toolkit, a single window opens with four tabs. Each tab is an independent module and can be used without touching the others.

Tab	Icon / Colour	What you do here
GSTIN Validator	Blue	Paste a GSTIN, validate it, get corrected suggestions.
HSN / SAC Search	Green	Look up HSN or SAC by code or by keyword.
GSTR-1 JSON Check	Orange	Validate & rectify a GSTR-1 JSON; export to Excel.
GSTR-2B Reconciliation	Purple	Match books (purchase register) against GSTR-2B.

✓ Recommended first-time workflow: open the HSN/SAC tab and load the master file once. This populates hsn_cache.json so that every other module can validate HSN/SAC instantly.

4 GSTIN Validator

A GSTIN (Goods and Services Tax Identification Number) is a 15-character identifier with the following structure:

```
[1-2]   [3-7]     [8-11]   [12]      [13]   [14]   [15]
State   PAN Alpha  PAN Num  PAN Alpha Entity Z      Check
 code   (5)        (4)      (1)       (1)          digit
```

Validation logic

The toolkit validates a GSTIN in two steps:

- **Structural check** — regular-expression match for the 15-character pattern.
- **Checksum check** — the official **Mod-36 (Luhn variant)** algorithm recomputes the 15th character from the first 14 and compares it.

How to use the tab

- Type or paste a 15-character GSTIN into the input box.
- Click **Validate** — the result appears below in green (valid) or red (invalid).
- If the checksum fails, two correction panels open automatically:

Panel	Assumption	What it shows
Fix #1	First 14 characters are correct	The single correct 15th character.
Fix #2	The 15th character is correct	Every 14-character alternative (one position replaced) that would produce a matching checksum.

> * Double-click any row in the Fix #2 table to copy the corrected GSTIN straight to the clipboard.

Common pitfalls

- Trailing spaces break the checksum — always paste clean text.
- Lowercase letters are handled automatically (auto-uppercased).
- A GSTIN can be structurally valid but have a wrong checksum — that combination usually means a transcription error in one character.

5 HSN / SAC Search

The HSN/SAC tab provides a two-way search against the master data you supply. HSN covers goods; SAC covers services.

First-time setup

- Click **Load HSN/SAC Excel...** at the top of the tab.
- Select your *HSN_SAC (3).xlsx* file.
- The application creates **hsn_cache.json** next to the script so subsequent launches are instantaneous.

Two ways to search

Method	Example query	Result
By code	10063010	Exact HSN entry (rice, semi-milled / wholly milled)
By description	basmati rice	Ranked fuzzy matches with similarity score
Partial code	1006	All HSN codes starting with 1006 (chapter heading)

> i Use the 'Dataset' dropdown to restrict results to HSN, SAC, or both.

Selecting and copying

- Click any row to highlight it; the exact code appears in the **Selected** field at the bottom.
- Click **Copy Selected Code** to place it on the clipboard.

6 GSTR-1 JSON Checker

This module loads a GSTR-1 JSON file, verifies every supplier GSTIN and every item HSN/SAC code before you upload it to the portal, lets you stage corrections, and exports the fixed data to JSON or Excel.

Workflow

Step	Action	Description
1	Load JSON	Click **Load GSTR-1 JSON...** and pick your file. The tab shows GSTIN and filing period from the JSON.
2	Scan	Click **Validate GSTIN & HSN**. Every issue appears as a row: type (GSTIN/HSN), JSON path, current value, message.
3	Inspect suggestions	Select a row. For a wrong GSTIN you get the Mod-36 corrected candidates. For a wrong HSN you get fuzzy matches from the master.
4	Stage a fix	Type the corrected value in the input box and click **Validate & Stage**. The value is re-validated; invalid values are refused.
5	Export	Click **Apply Corrections & Export JSON** to save a *_corrected.json, or **Export JSON** → **Excel** for a spreadsheet.

What is validated

- **Supplier GSTINs** — every *b2b[].ctin* and *b2ba[].ctin*.
- **Item HSN/SAC codes** — every *itms[].itm_det.hsn_sc / hsn / hsn_code*.
- **HSN summary rows** — the optional *hsn.data[].hsn_sc* array.

Excel output — sheet by sheet

Sheet	Contents
B2B Invoices	One row per invoice, with totals of taxable value & tax.
B2BA	Amendments to prior-period invoices.
HSN Summary	HSN-wise taxable value, IGST, CGST, SGST, Cess.
B2CS	Business-to-consumer supplies summarised.

i The Excel file auto-opens after saving (Windows). On other platforms the file is saved but you must open it manually.

7 GSTR-2B Reconciliation

! Only the sheet named exactly 'B2B' inside the GSTR-2B workbook is read. Other sheets (B2BA, CDNR, IMPG, etc.) are ignored by design.

Four-step workflow

Step	What you do	Options
1	Select the Books / Purchase Register file.	Select existing file, or **Create Books Template** to generate a blank one.
2	Select the GSTR-2B file(s).	Single file, or multiple files that will be merged (B2B rows concatenated without duplicate headings).
3	Set tax tolerance and click **Run Reconciliation**.	Tolerance is in rupees — default ■1 per tax head.
4	Export the result.	Excel, Word or PDF.

Books template — column layout

```
GSTN | PARTY NAME | VOUCHER TYPE | INVOICE NO | INVOICE DATE |
TAXABLE VALUE | IGST | CGST | SGST | INVOICE VALUE
```

Categories in the reconciliation output

Category	Meaning
Matched	The invoice exists on both sides within tolerance.
Amount Mismatch	Same GSTIN + Invoice No, but tax amounts differ beyond tolerance.
Only in Books	Invoice found in the books but absent from GSTR-2B.
Only in 2B	Invoice reported by supplier in GSTR-2B but missing from books.
GSTIN Invalid (Books)	Books row with an invalid GSTIN (Mod-36 check failed).
GSTIN Invalid (2B)	GSTR-2B row with an invalid GSTIN.

Matching logic

- **Primary key** — (GSTIN, Invoice Number). Invoice numbers are normalised (trimmed, uppercased, trailing .0 stripped).
- **Tolerance check** — sum of absolute differences across IGST, CGST and SGST must be ≤ the configured tolerance.
- Each row is consumed only once — no double matching.

8 Exports & Reports

Every module produces outputs you can hand over or archive. Below is a consolidated view of what each export delivers.

Source	Format	Content
GSTR-1 Checker	JSON	Corrected copy of the original GSTR-1 JSON.
GSTR-1 Checker	Excel	Sheets: B2B Invoices · B2BA · HSN Summary · B2CS.
GSTR-2B Reconciliation	Excel	One sheet per category with all columns.
GSTR-2B Reconciliation	Word	Formatted report with headings and tables.
GSTR-2B Reconciliation	PDF	Landscape A4 report, ready to email to clients.

✓ All exports auto-open in Windows after saving, and never overwrite an existing file without asking for confirmation.

9 Tips, Do's & Don'ts

Do

- Load the HSN/SAC master **once** and rely on the cache thereafter.
- Always keep a backup of the original JSON before rectifying.
- Use the multi-file option to merge GSTR-2B files for a full quarter.
- Set the tolerance to ■**1** initially; widen it only if you know suppliers round differently.
- Export to Word or PDF for archival and audit trails.

Don't

- Don't paste GSTINs with hidden spaces or newlines.
- Don't rename the sheet named 'B2B' in a GSTR-2B workbook — the reader looks for it by exact name.
- Don't edit the generated hsn_cache.json manually.
- Don't assume 2B is complete: reconcile before claiming ITC.

* A 2-minute daily routine: open the toolkit, load HSN master if not already cached, then validate the day's GSTR-1 JSON before upload.

10 Troubleshooting

Symptom	Likely cause	Fix
Valid GSTIN rejected	Old buggy checksum formula (pre-v1.0.1)	Ensure *gstin_check_digit* uses **(36 - total % 36) % 36**.
HSN tab empty	Master never loaded	Click **Load HSN/SAC Excel...** once.
2B reconciliation shows 0 rows	Sheet not named exactly 'B2B'	Rename the sheet to **B2B**.
Books file rejected	Column headers differ	Use the built-in **Create Books Template**.
PDF export blank	No recon run yet	Run reconciliation before exporting.
Import error on launch	Missing package	Re-run the pip install command from Section 2.

! When reporting an issue, attach a screenshot of the error dialog and the exact text shown in the status bar.

11 Appendix: Glossary

Term	Meaning
GSTIN	15-character Goods and Services Tax Identification Number.
HSN	Harmonised System of Nomenclature — classification for goods.
SAC	Services Accounting Code — classification for services.
Mod-36	The Luhn mod-36 checksum algorithm used by GSTN.
GSTR-1	Outward supplies return filed by a registered person.
GSTR-2B	Auto-drafted ITC statement generated from supplier filings.
ITC	Input Tax Credit — credit available on inward supplies.
B2B	Business-to-business supply.
B2CS	Business-to-consumer supplies, summarised.
IMS	Invoice Management System on the GST portal.
PAN	Permanent Account Number — embedded in GSTIN positions 3–12.
Tolerance	The rupee threshold below which tax differences are considered a match.

End of Manual

Thank you for using the GST Toolkit. For feature requests or bug reports, contact the developer who provided this manual.